
A Governed Multi-Agent Layer for a Power-to-X Pre-Feasibility Optimizer

Presented by:

EL OUKILI Tarik

Defended on **September 28, 2026**, in front of the jury composed of:

Pr. TAMTAOUI Ahmed

Pr. TOUNSI Karima

Pr. BAINA Amine

Mr. GRISARD Malo

Mr. MARTIN Paul

Mr. EL MAJIDI Hassan

- INPT (Examiner)

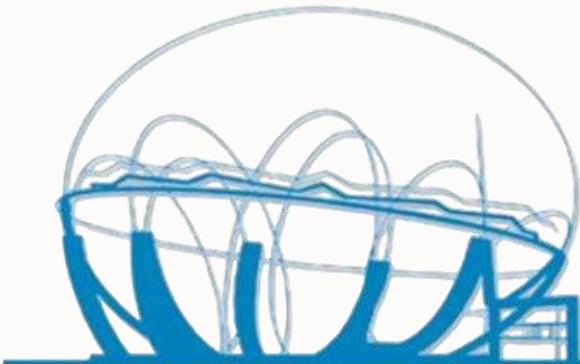
- INPT (Examiner)

- INPT (Internal Supervisor)

- BCG (Team Lead and Reviewer)

- BCG (Senior Team Member and Reviewer)

- BCG (External Supervisor)



Context

Problem

Manual workflow

Architecture

Implementation

Results

Conclusion

Agenda

1

Context

BCG X, the product,
the timeline

2

Problem

Expert-only tool, one
central question

3

Manual workflow

How a study is built by
hand today

4

Architecture

Authority flows one
way

5

Implementation

The delivered product
and operations

6

Results

Measured fidelity and
analyst effort

7

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Recognition and what
comes next

Section 1

Context

The host, the product, and the ground it stands on

BCG and BCG X: where the project was carried out



- **Founded in 1963** by Bruce Henderson, headquartered in Boston.
- **A reference in strategy consulting**, business transformation, and analytical problem solving.
- **The offering now integrates** data, technology, and digital-transformation services.



- **Accelerates digital transformation** by combining engineering, data, and design with the firm's strategy work.
- **Not a software vendor**: it helps organizations build capabilities and adopt new technologies.
- **Led by Sylvain Duranton**; regional centers include Morocco (Casablanca), Germany, China, Singapore, India.

—
1963

Founded

—
\$14.4B

Annual revenue, 2025

—
34K

Employees

—
100+

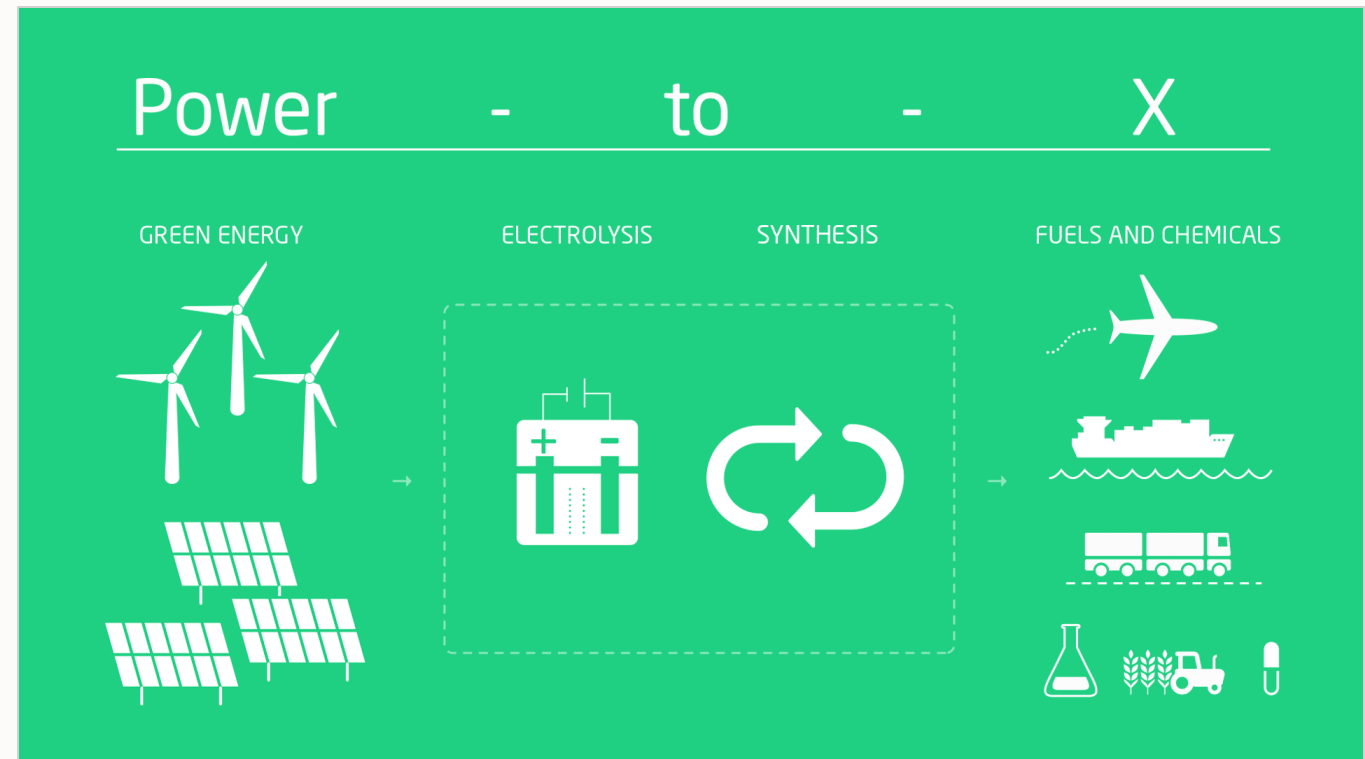
Offices worldwide

BCG X build areas

- AI and generative AI
- Customer experience for growth
- Venture and business builds
- Digital platform builds

Why Power-to-X: electrons where possible, molecules where not

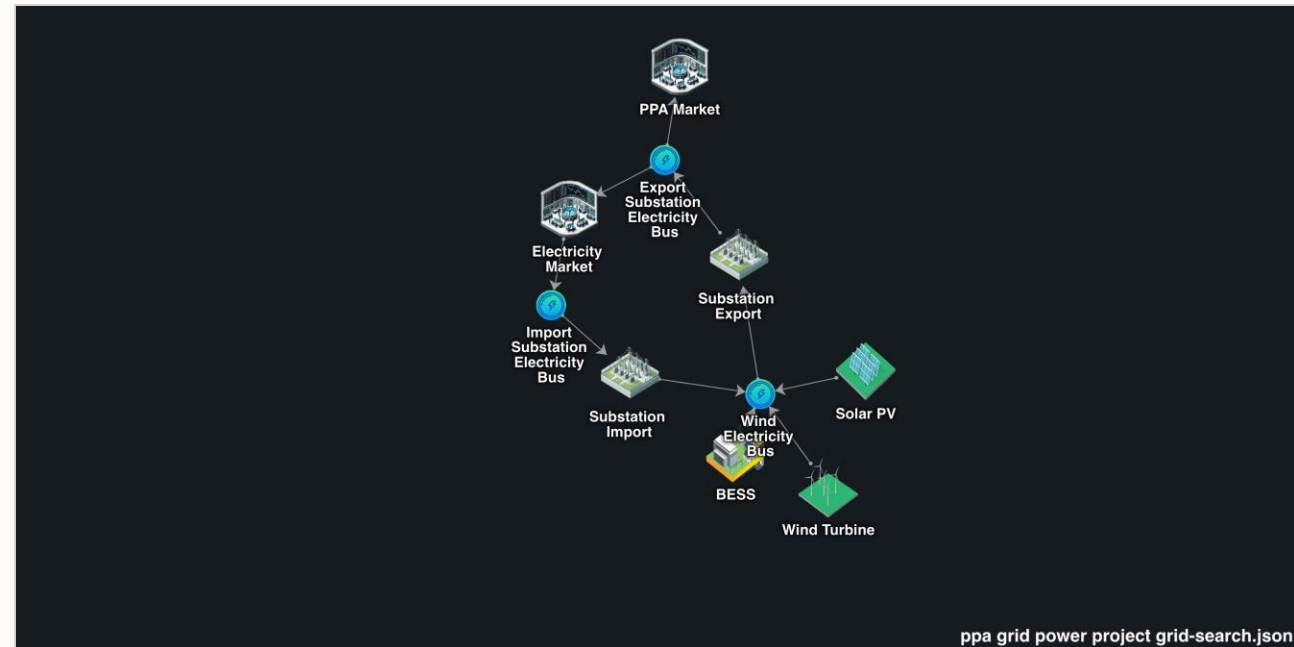
- Heavy industry, aviation, maritime: hard to electrify directly.
- **Power-to-X:** renewable electricity converted into storable carriers (hydrogen, ammonia, methanol, sustainable aviation fuel).
- **The catch:** projects are capital-intensive; pre-feasibility decisions rest on hundreds of uncertain technical and financial assumptions.



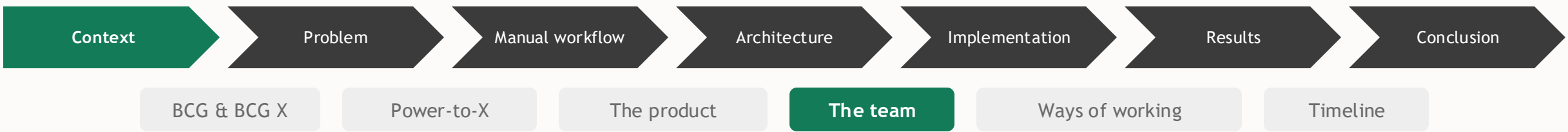
The Energy Transition Optimizer: a deterministic core

Energy Transition Optimizer by BCG X

- **A project is a typed graph:** assets, conservation nodes, markets, plus hundreds of parameters.
- **The deterministic engine** simulates operation hour by hour over the project lifetime.
- **Outputs:** auditable indicators such as NPV, IRR, and LCOX.
- **Keyword: deterministic.** Same inputs, same numbers, every time. Hold that thought.



An example project topology: assets, buses, and markets



The Energy Transition Optimizers team

- **A BCG X product team** dedicated to energy-transition analytics, partly operating from the Casablanca center with contributors across geographies.
- **A small product portfolio:** the Energy Transition Optimizer (project-level, the subject of this work), a Portfolio Optimizer, and a system-level Just Energy Transition tool.
- **Internal-product model:** build once, adapt per client engagement.

Member	Role	Main involvement
Malo Grisard	Product owner and technical lead	Product direction, prioritization, technical steering, and review.
Ariane Dalens	Product and technical contributor	Product context, review, and alignment with the product team's expectations.
Paul Martin	Product and technical contributor	Product context, technical feedback, and contribution to the delivery environment.
Mohamed Chadi Taqi	Software engineering	Platform recovery, frontend renovation, agentic core, and platform operations.
Ayman Youss	Software engineering	Simulation-log explanation, hardening, and evaluation-harness support.
Tarik El Oukili	Data and AI engineering	Result-interpretation agent, results-visualization layer, LLM-as-a-judge evaluation, supporting read-only tools, and late-stage stabilization.

Ways of working: BCG engagement cadence

- **Agile delivery**, adapted to a product-modernization effort: short cycles and overlapping thematic phases, with continuous integration keeping the application deployable throughout.
- **Daily check-in and check-out** to align on priorities and hand off open items.
- **Case Team Meeting** once a week with the engagement Managing Directors and Partners: direction, risks, key decisions.
- **Monthly retrospective** on what worked and what to improve.



Zoom



Teams

Teams



Trello



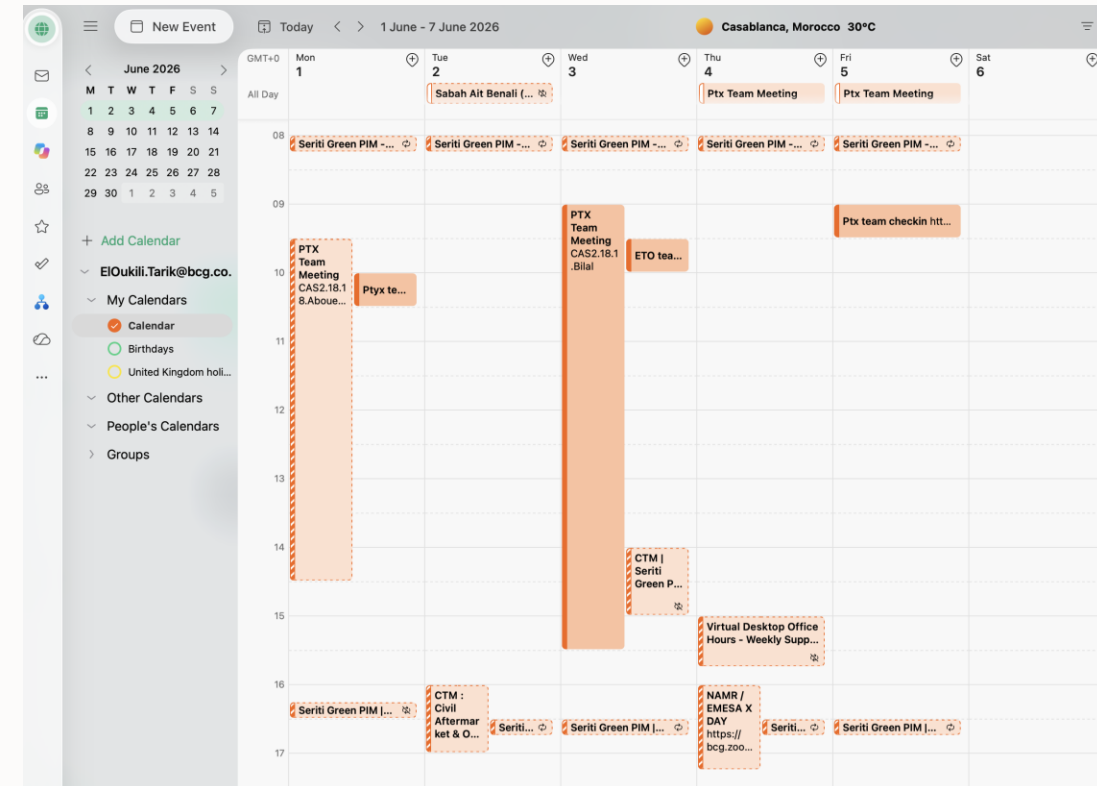
Slack



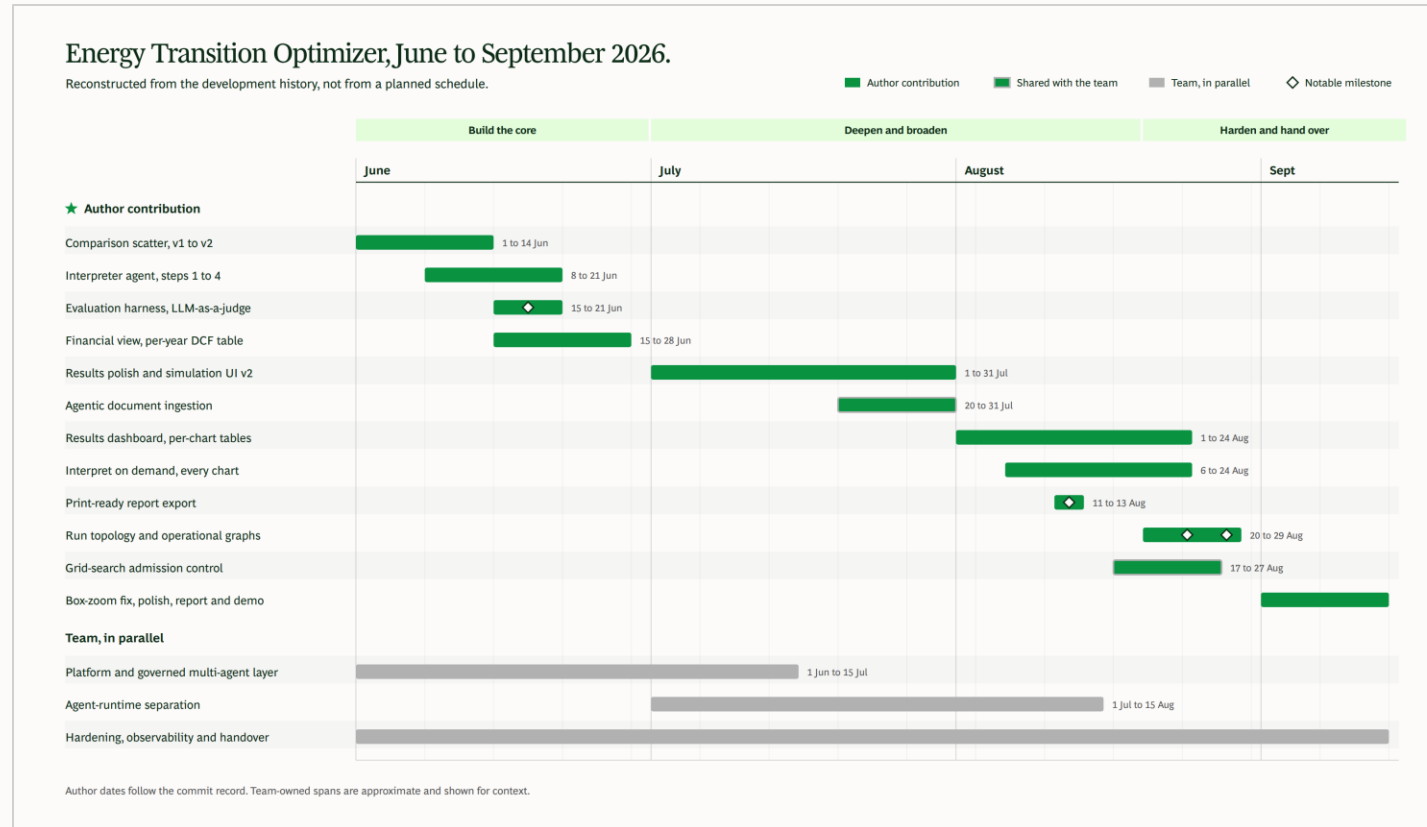
Outlook



GitHub



The internship: four months, overlapping phases



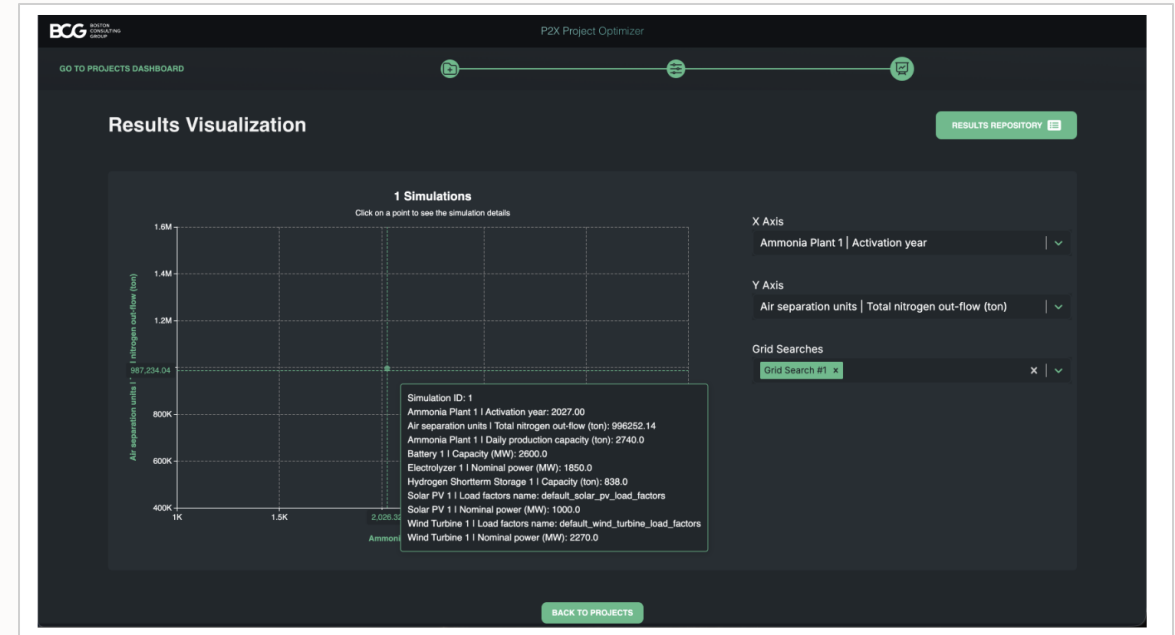
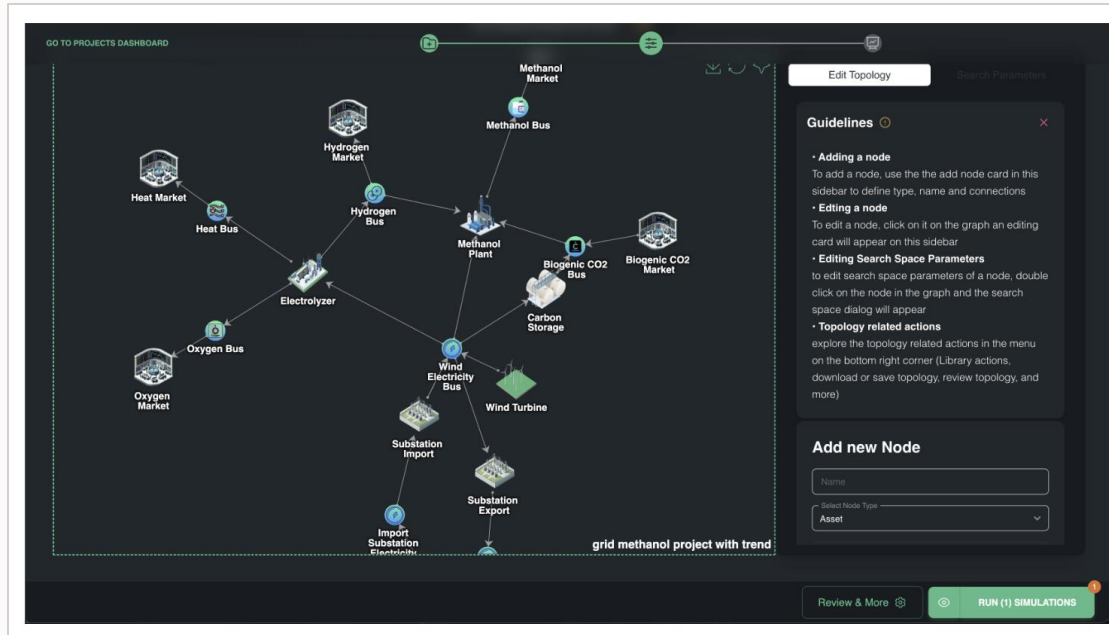
- **June to September 2026:** platform foundation first, then the governed agentic core, then the specialist capabilities, then observability and hardening.
- **The specialist capabilities matured in the same arc:** visualization, then interpretation, then evaluation.

Section 2

Problem

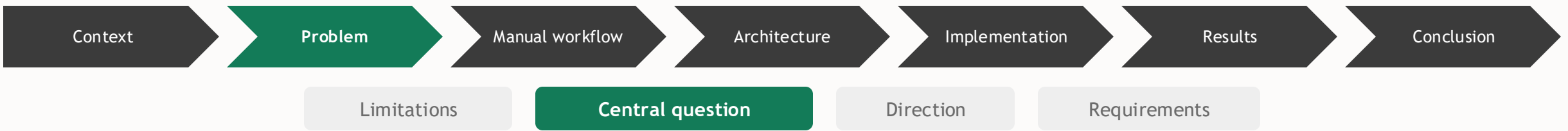
An expert-only product on a fragile platform, and one central question

A powerful product with two problems



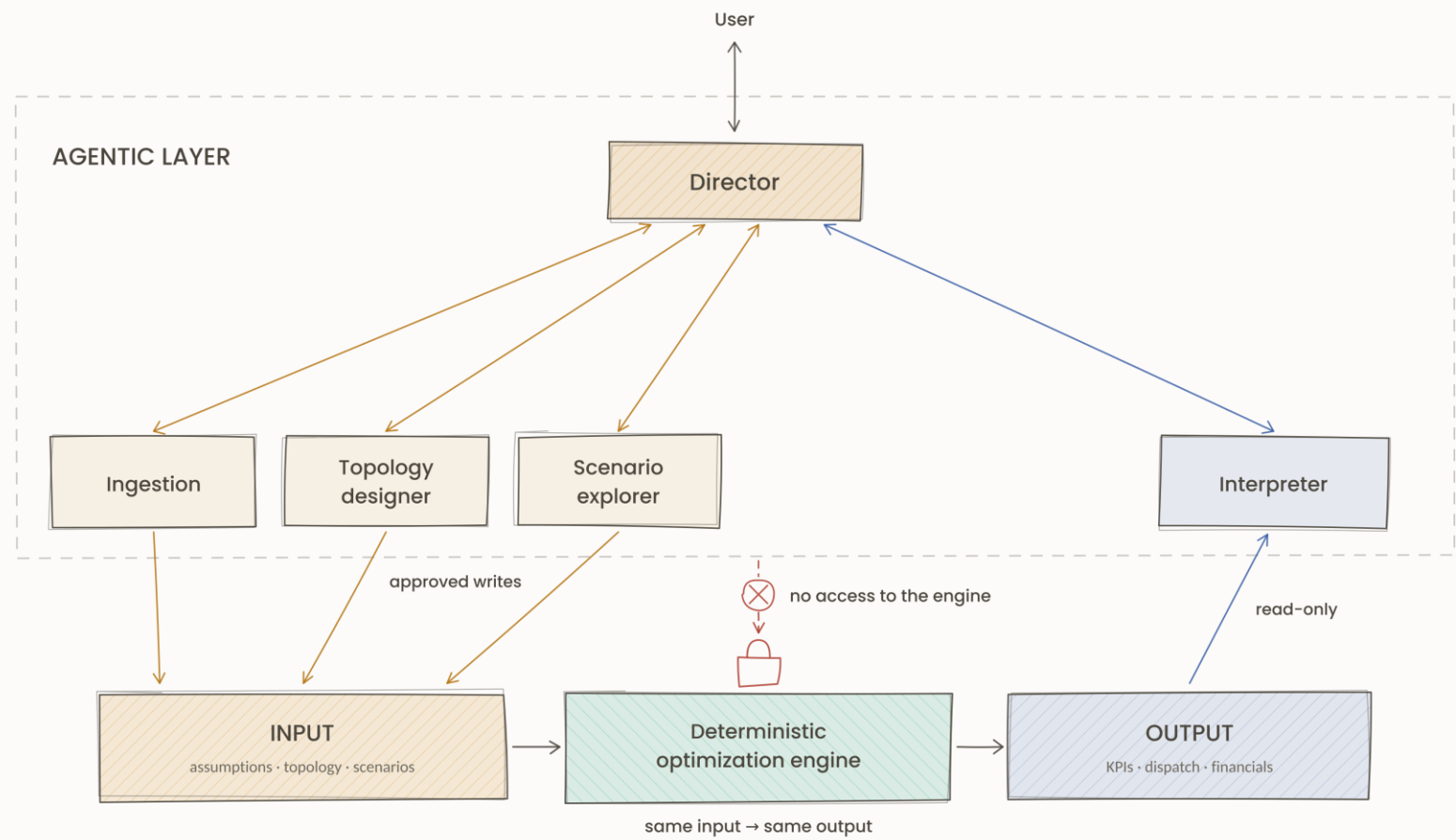
The inherited, expert-only interface

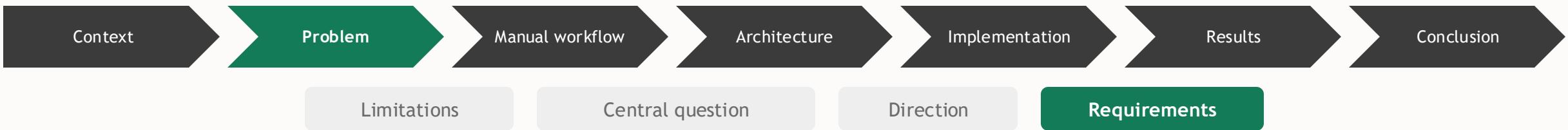
1. **Fragile platform:** broken delivery pipeline, identity security findings, infrastructure drift, weak observability
2. **Expert-only:** hand-built topologies, hundreds of parameters, raw result populations to read



How can generative AI make an expert-only techno-economic optimizer accessible in natural language, without compromising the reproducibility, auditability, and human control its results demand?

Chosen direction: a governed multi-agent layer





Requirements: what had to be delivered end to end

Ref.	Requirement	Priority
F6	Governed state changes	Must
F4	Result interpretation	Must
F5	Results visualization	Must
F7	Evaluation of behaviour	Must
F1	Guided topology design	Should
F2	Assumption ingestion and audit	Should
F3	Scenario exploration	Should
F8	Observability and monitoring	Should

- **Must:** governed state changes (F6), result interpretation (F4), visualization (F5), evaluation (F7).
- **Should:** capabilities largely in place when this phase began (topology design, ingestion, exploration, observability).
- **Non-functional backbone:** groundedness, auditability, human control, measured quality.

Section 3

The manual workflow

How a study is built by hand today, and what it costs

Step 1: introduce the assumptions, by hand

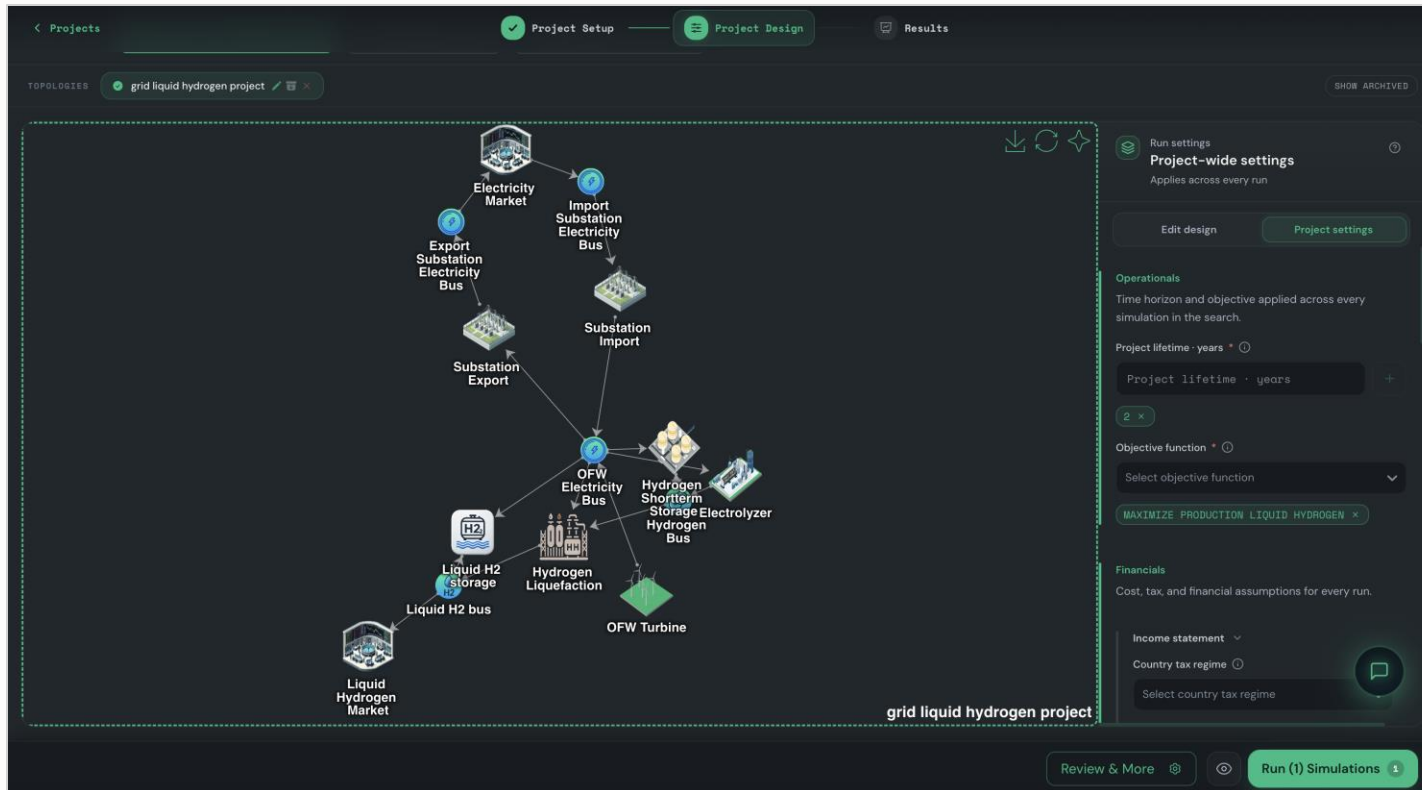
The screenshot displays the 'Energy Large Capital Projects' software interface. The top navigation bar includes 'Projects', 'Project Setup', 'Project Design', and 'Results'. The current step is 'STEP 01 · PROJECT DATA', with a sub-header 'Define project data' and a description: 'Start with preset data, or upload your own files by category.' Below this, there are tabs for 'Preset data' and 'Custom data'. The main content area is titled 'Markets' and lists six market categories: Ammonia market, Biomass market, Carbon feedstock market, Diesel market, DRI market, and Electricity leeway downregulation market. Each category has a 'Preset' button and a 'Manage' button. A sidebar on the left shows 'CATEGORIES' with 'Assets', 'Financials', 'Markets', 'Load Factors', and 'Performance C...'. A 'GUIDE' section on the right provides instructions on how to use the data. At the bottom right, there is a 'Validate' button.

- **Every assumption is bound by hand** to its node: technical, commercial, financial, tax.
- **Hundreds of parameters per study**, spread across screens and spreadsheets.

Project data: assets, markets, financing, taxes, and performance inputs

Average manual effort: about 3 hours per study

Step 2: design the topology, by hand

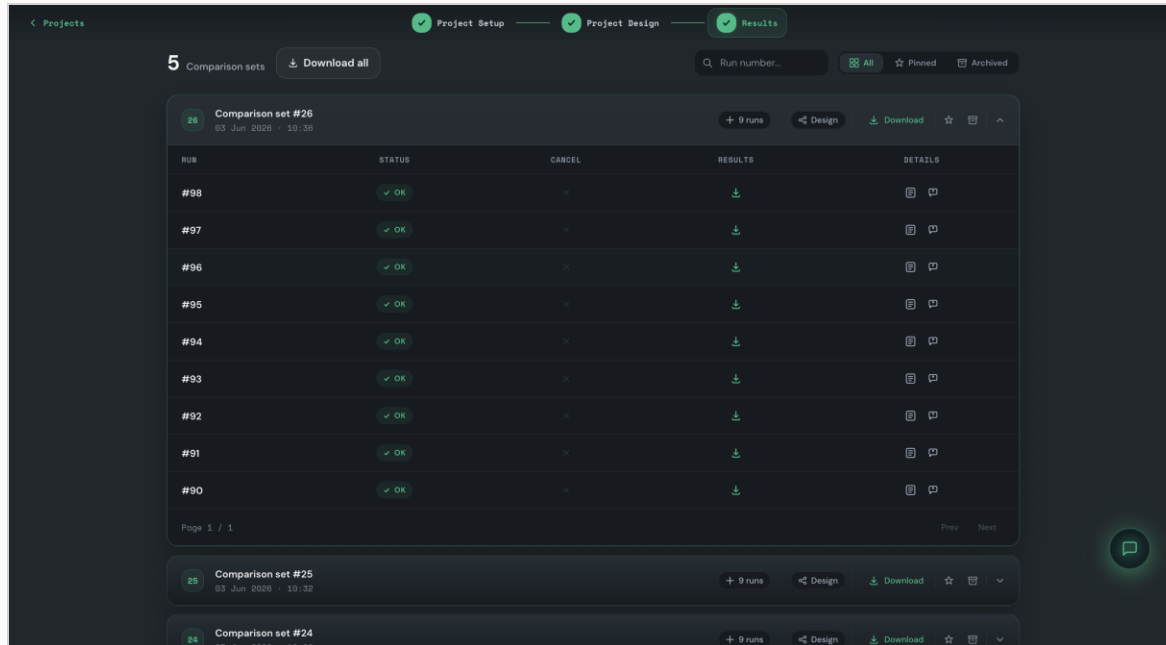


The topology workspace: the project graph, built and edited manually

- A **valid typed graph** of assets, conservation nodes, and markets must be assembled by hand.
- **Component-level parameters** set one by one for the engine to consume.

Average manual effort: about 4 hours per study

Step 3: run the sweep and read the results, on your own



Raw outputs: dense comparison tables and per-run charts, expert reading required

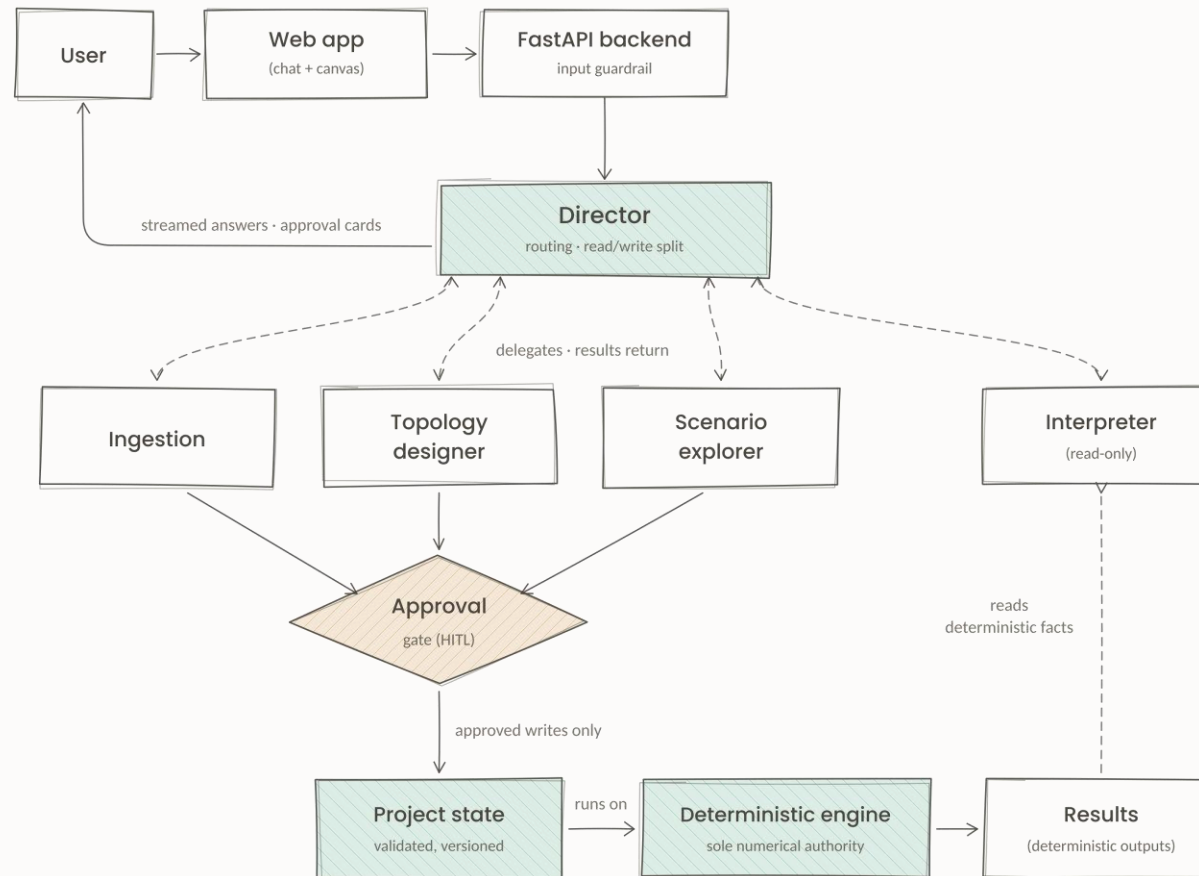
About 6 hours to configure a grid search, plus about 2 hours per expert interpretation

Section 4

Architecture

Authority flows one way

System architecture: authority flows one way



dashed = delegation and read-only reads · solid = data and state flow
approval commits state; the engine computes from it

Implementation

The delivered product and its operations

Context

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Conclusion

Stack

Visualization

Agents

Platform

Built with a pragmatic, production stack

Frontend



Next.js



TypeScript

Backend and deterministic execution



Python



FastAPI



OR-Tools

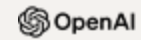


Celery

Agentic



LangChain/Graph



OpenAI models



Langfuse

Persistence and messaging



PostgreSQL



Redis

Cloud, identity, and runtime



Azure



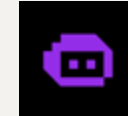
Terraform



Docker



Kubernetes



Kagent

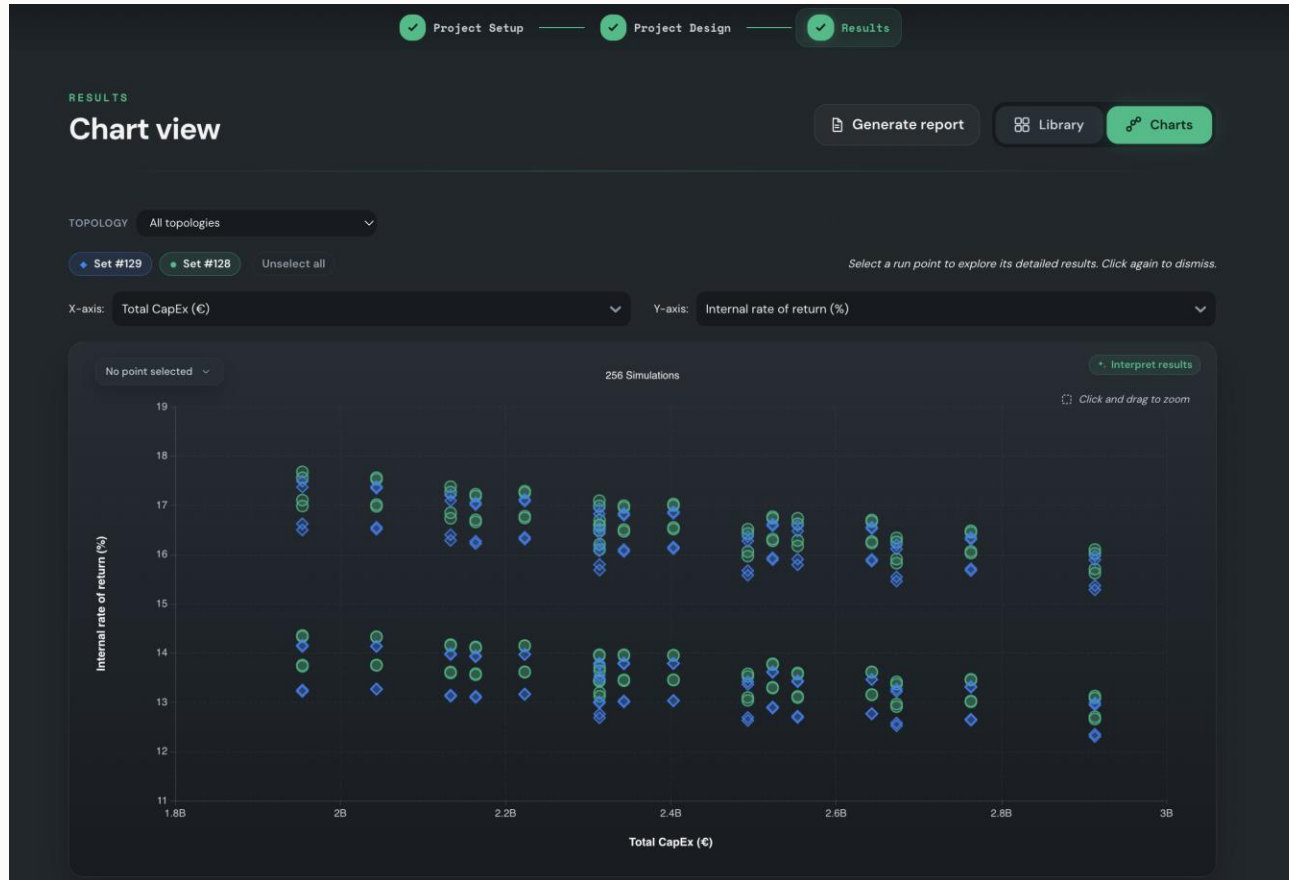


Okta



CircleCI

Visualization: from a population to one run



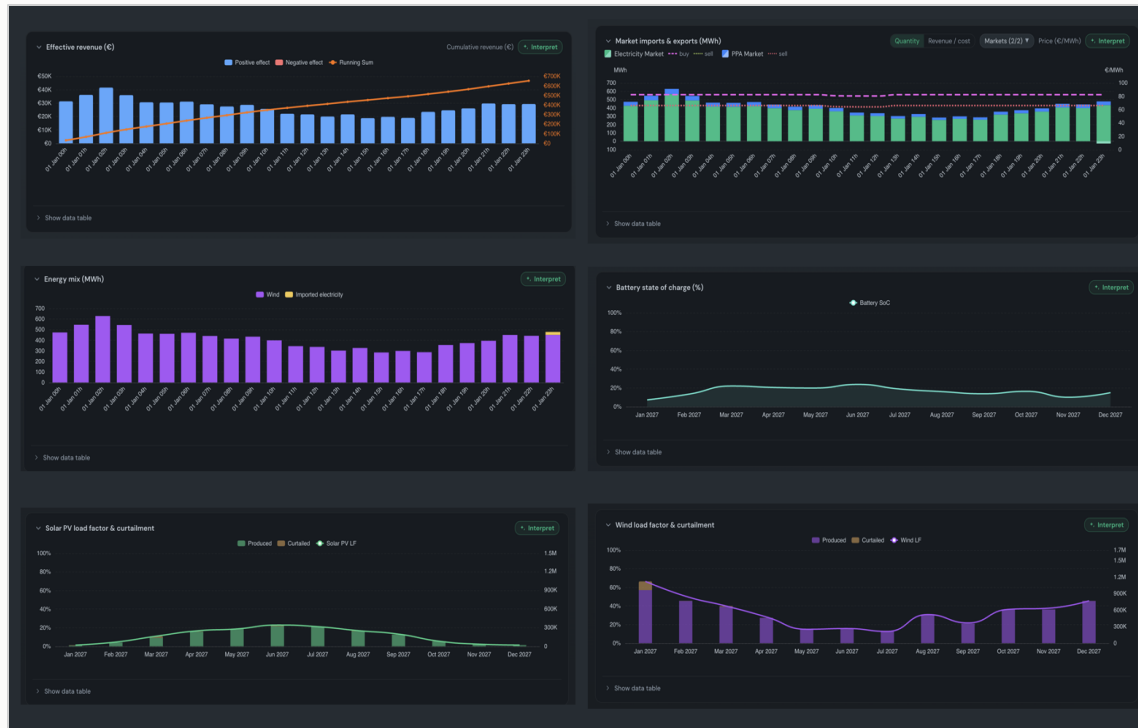
Reading the population

- One point per simulation across the whole population
- Selectable axes for side-by-side comparison
- Pick a point to open the single run behind it

Every chart carries an interpret-on-demand action

Comparison scatter with selectable axes

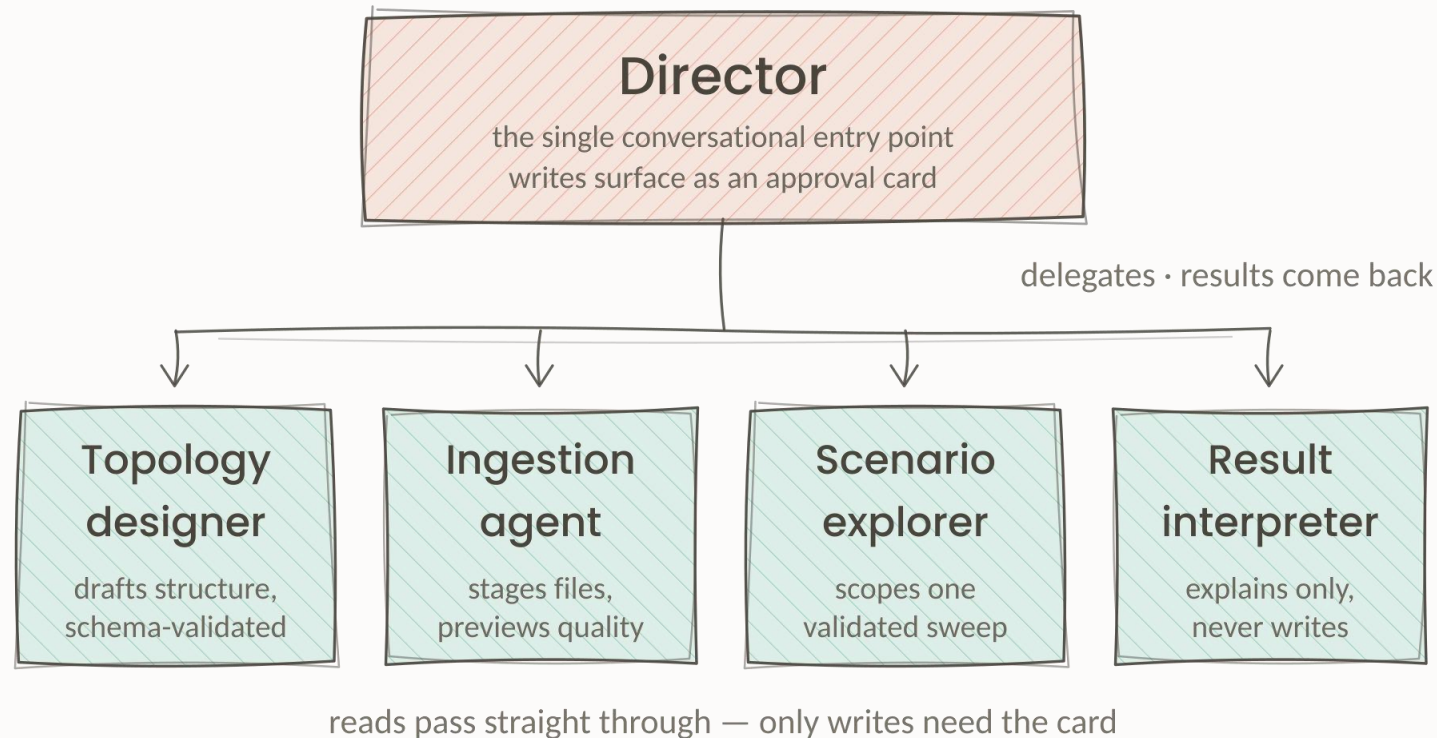
Visualization: the operational dashboard and the financial view



Six-panel operational dashboard (left) and navigable per-year DCF table (right)

Every chart carries an interpret-on-demand action

One governed layer: a Director delegating to specialists

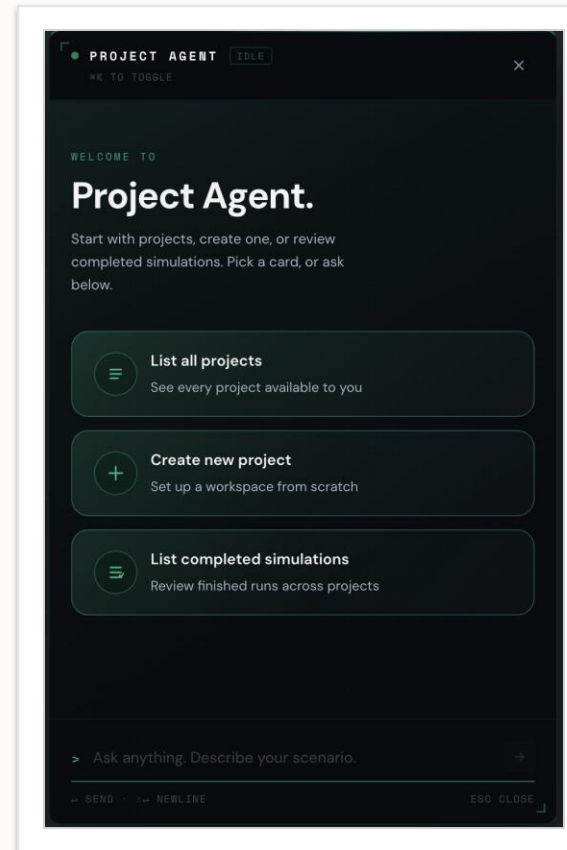


Every state-changing action, from any agent, passes the same gate: typed proposal, system validation, human approval

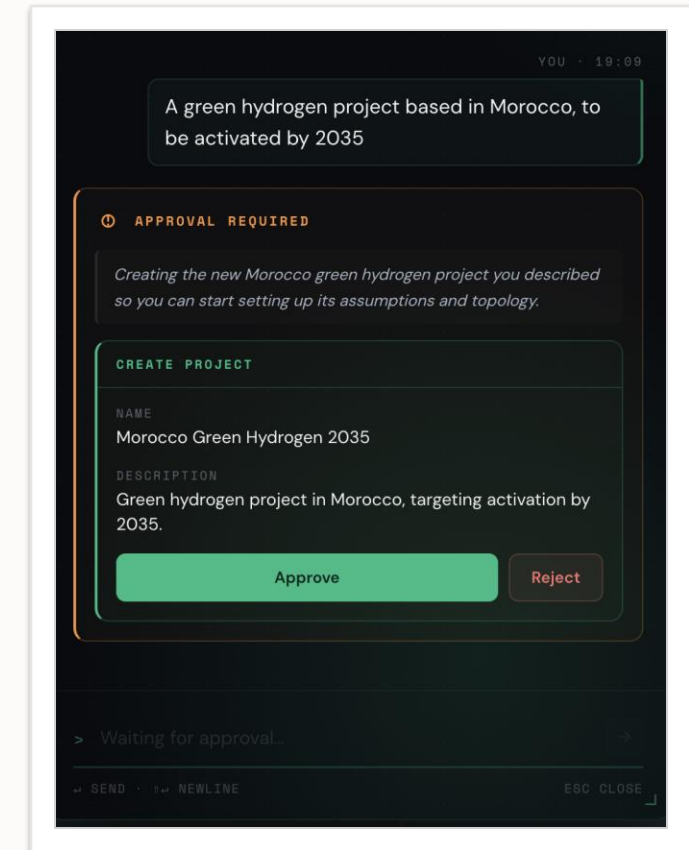
The Director: one conversational entry point

- **Grounds every request** in the active project, page, and conversation.
- **Answers reads directly;** write intent becomes an approval card.
- **Suppresses duplicate calls** in a turn, and stops a proposal being narrated as done.

Every write becomes an approval card; nothing changes without the user.



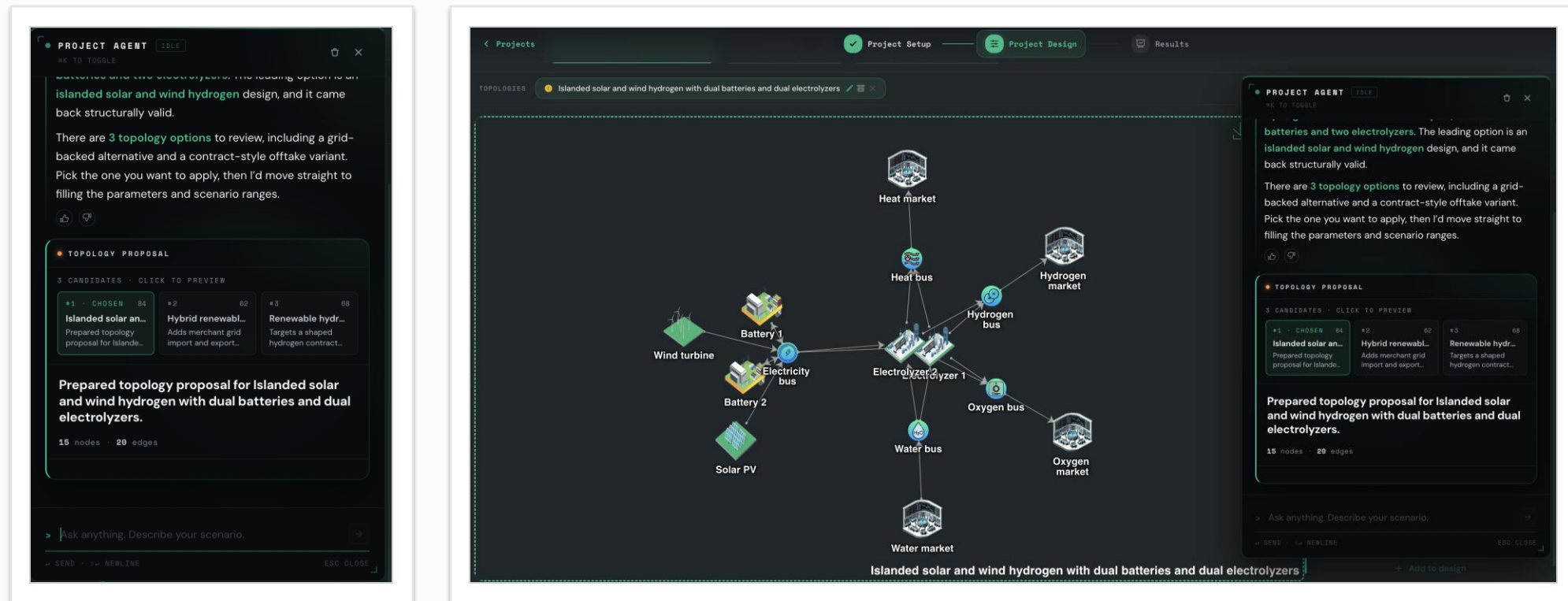
Guided entry points and a free-form prompt



A write intent, turned into an approval card

The Topology Designer: from intent to a validated draft

A validated design is still a draft; the canvas changes only on approval.

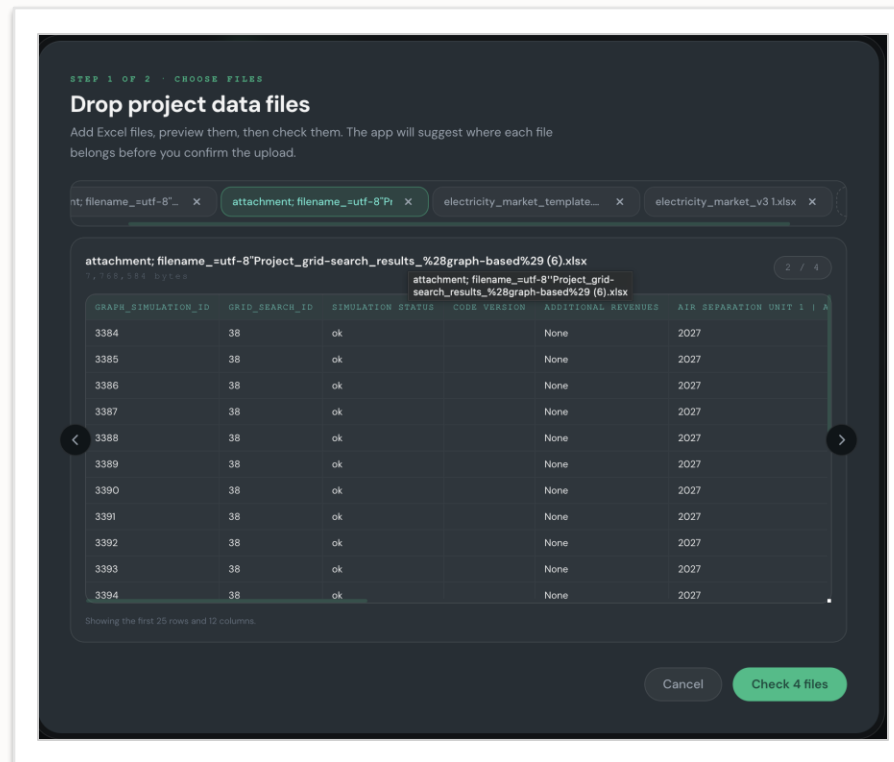


Candidate structures

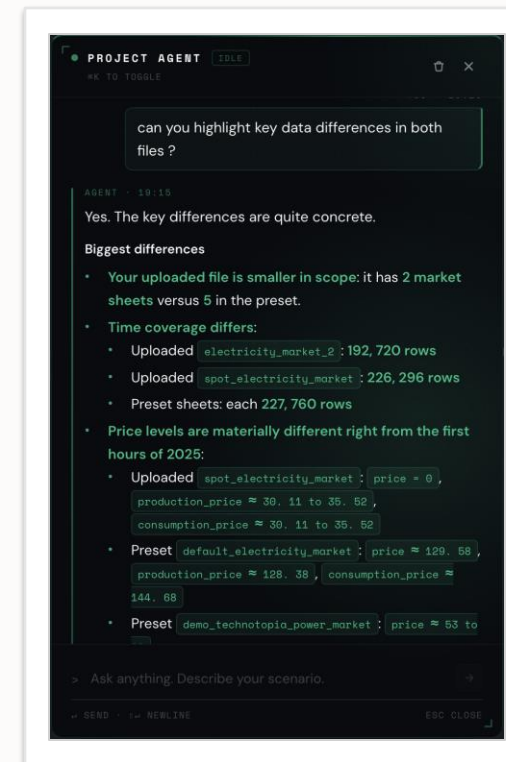
The approved structure on the canvas

The Ingestion Agent: staged, reviewable data intake

Uploads land as a preview; no project data changes until it is accepted.



Sheet selection with a parsed-row preview

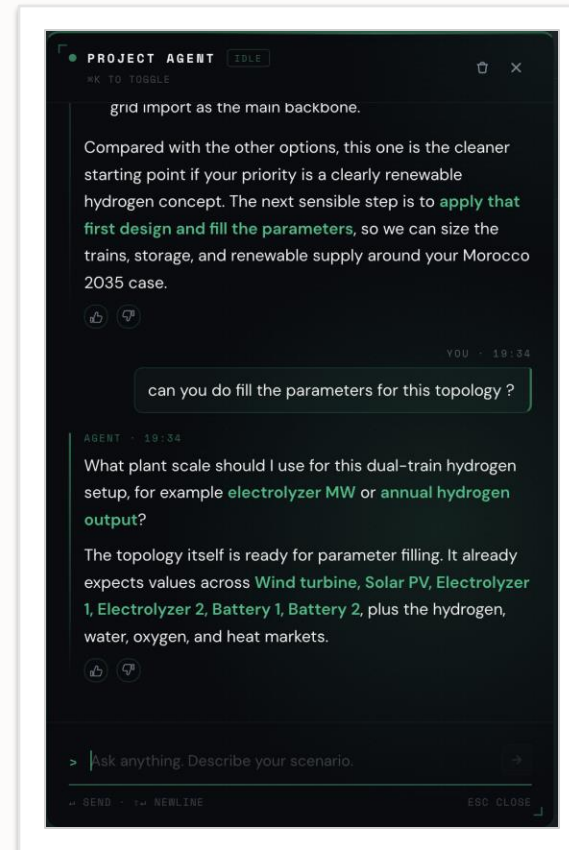


Differences made explicit

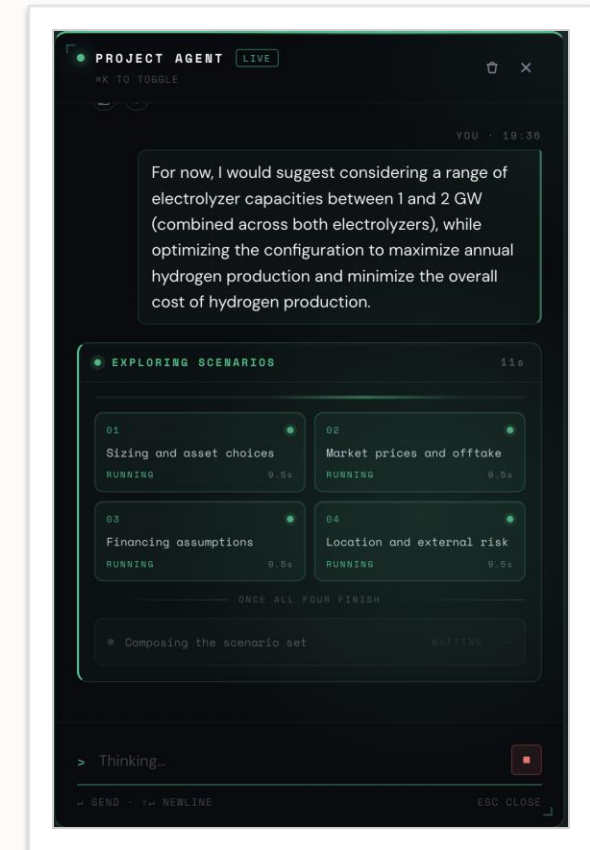
The Scenario Explorer: one analyst, four axis agents

- A **scoping Analyst** checks the request and asks rather than invents what is missing.
- **Four axis agents run in parallel:** technical, commercial, financial, external.
- **The Analyst reconciles them** into one validated set of canvas operations.

Four agents, one approval: the sweep lands as a single reviewed change.



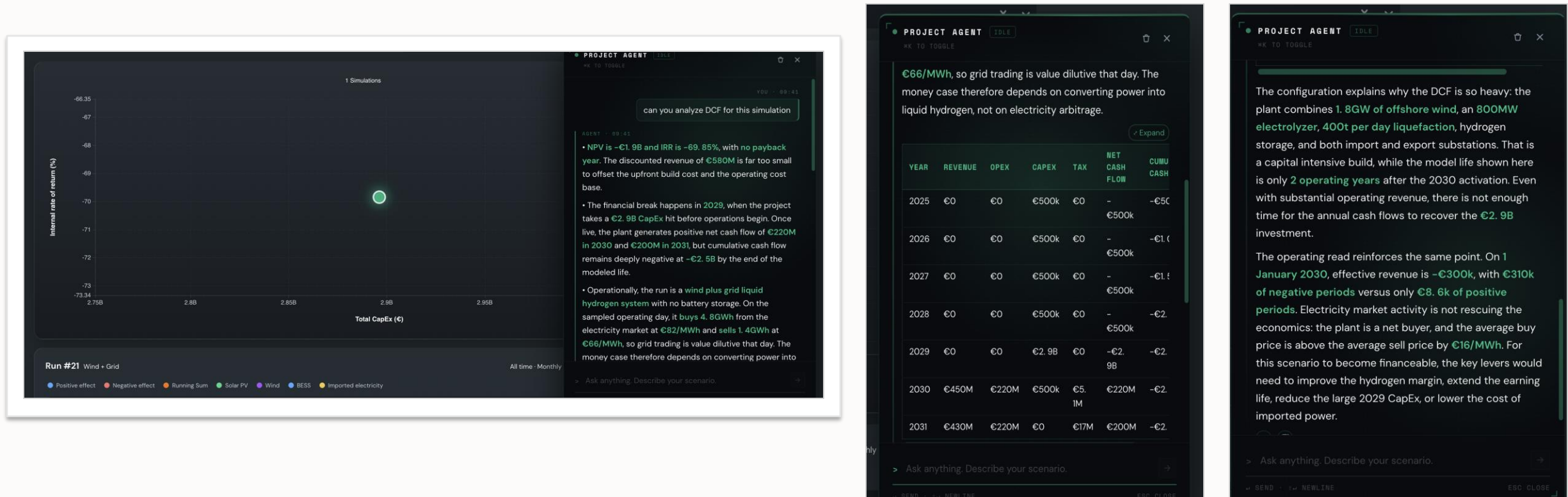
Scoping short-circuit: ask, not invent



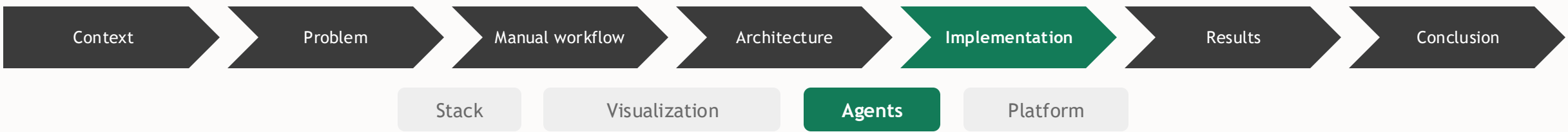
A manageable sweep from study directions

The Result Interpreter: grounded answers, read-only

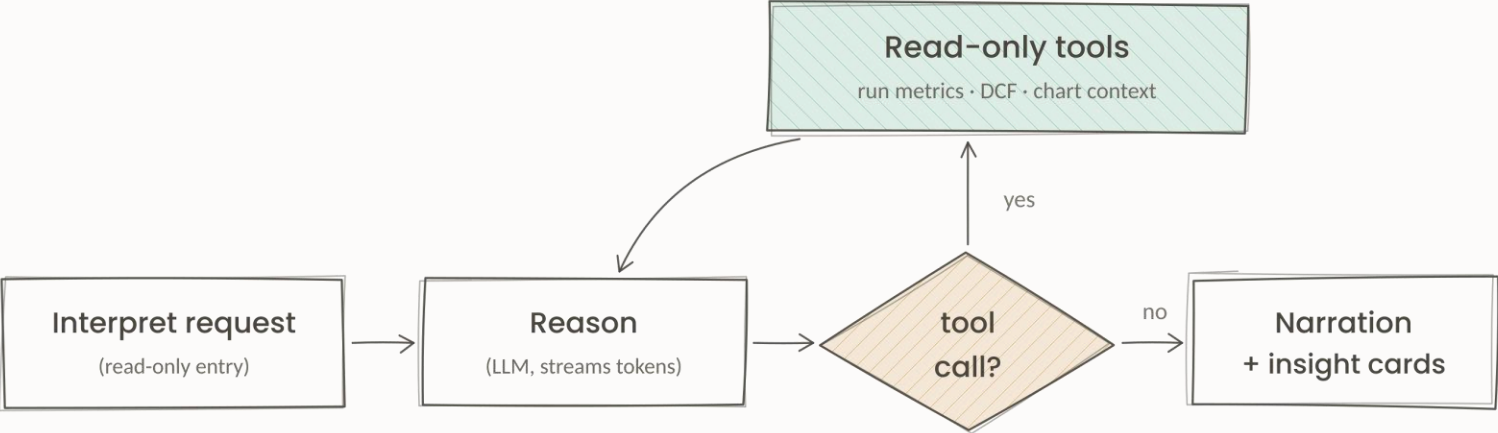
Read-only by construction: no approval gate, because nothing can mutate.



A run analyzed beside its scatter; in the chat, an embedded DCF table and a grounded narrative citing deterministic figures

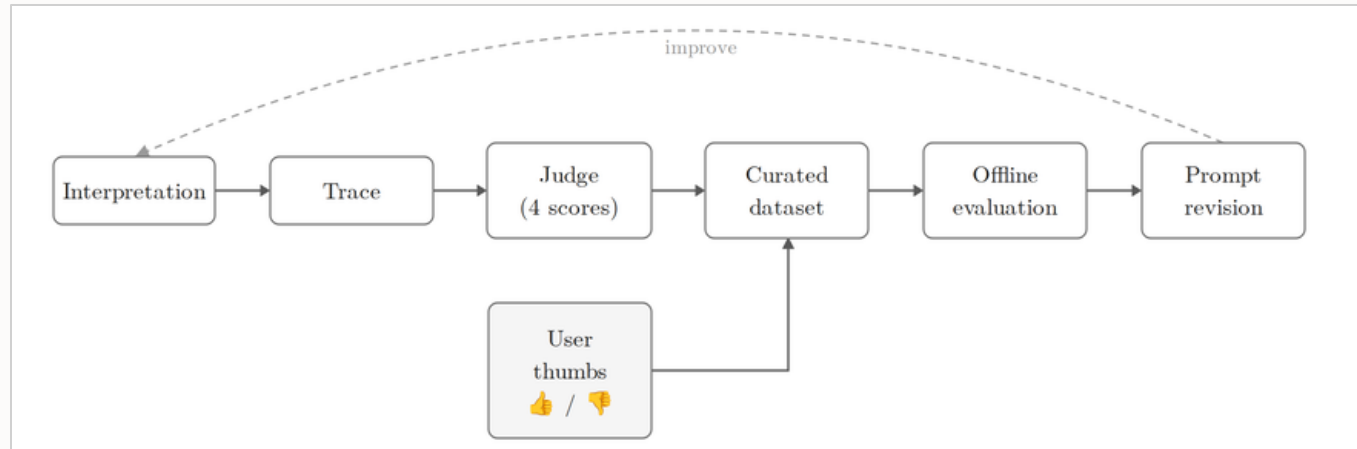


Result interpretation: grounded, read-only, bounded



- **Reason-and-act loop over read-only tools** that call the same unit-tested analytics that draw the charts.
- **The persona forbids computing figures:** every number cited must come from a tool result.
- **Bounded** tool rounds and a hard timeout; fresh stateless turn, compact pre-aggregated facts.
- **Restraint by design:** one point on a scatter gives “no relationship can be inferred”, not a fabricated trend.

Evaluation: measured, not assumed



- **Independent LLM judge**, outside the interpreter loop, scores each explanation 0 to 1 on four dimensions:

Groundedness: every figure traces to a tool fact

Financial soundness: discounting, cash flow, tax

Commercial soundness: realistic prices for the setup

Operational coherence: a causal story that hangs together

- **Online:** sampled, scores written to each trace.
- **Offline:** curated dataset, score deltas across versions.
- **User thumbs** promote weak turns into the dataset.

The judge at work: four scores on a real trace

agentic.chat.turn ID

+ Add to datasets

Annotate

▼

Add comment

2026-06-21 17:14:35.104

Latency: 17.41s

Env: testRelease: dev

142,713 prompt → 2,210 completion (Σ 76,992)

Preview

Log View

Scores

Show filters

Views 0

1dPast 1 day

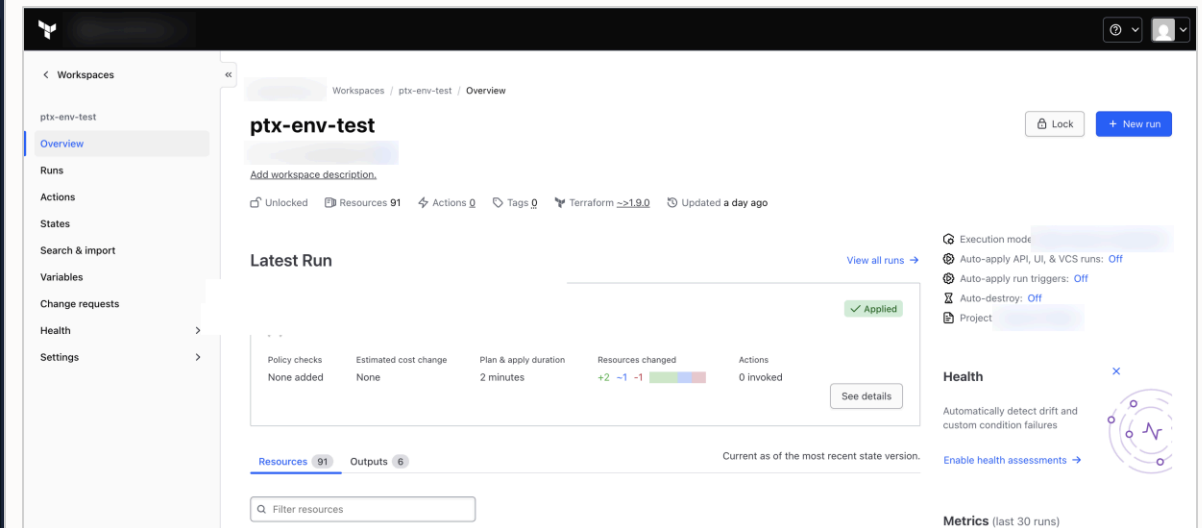
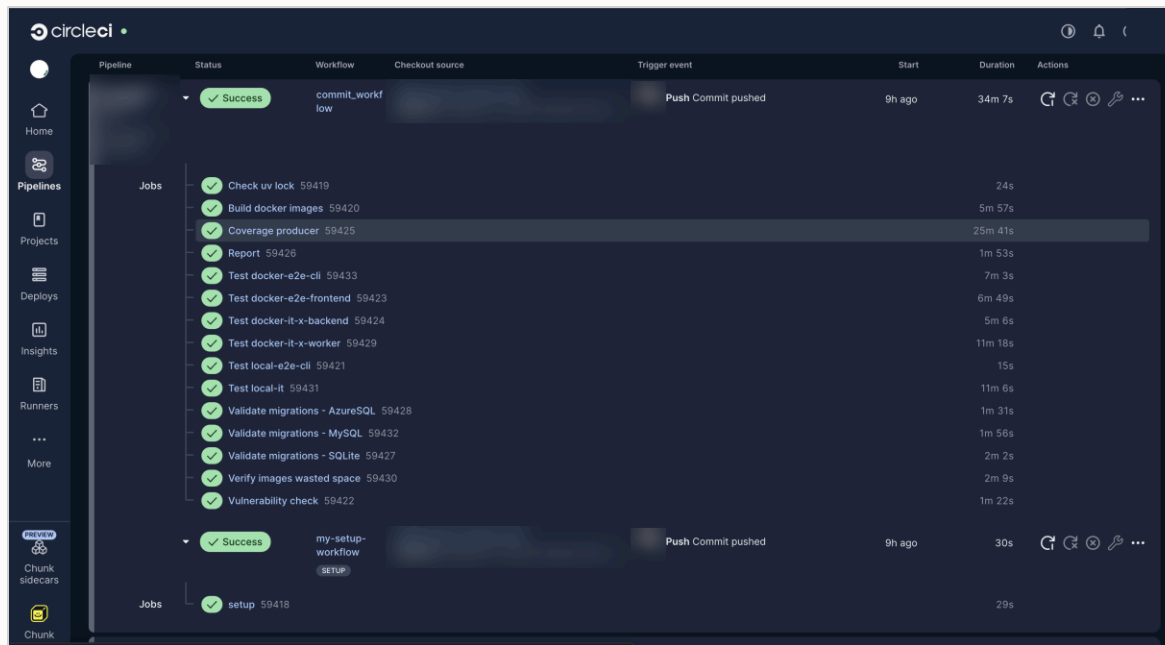
Columns 12/14

☐	Name	Data Type	Value	Metadata
☐	commercial-soundness	NUMERIC	0.9600	{}
☐	financial-soundness	NUMERIC	0.9400	{}
☐	groundedness	NUMERIC	1	{}
☐	operational-coherence	NUMERIC	0.9700	{}

Four judge dimensions written onto an interpretation's trace in Langfuse

Delivery: from code change to controlled deployment

- **Application changes** pass through continuous-integration workflows;
- **infrastructure changes** are applied as code, so every environment is reproducible.



The restored delivery chain: CircleCI workflows (dependency checks, builds, tests, migrations, scans) and Terraform runs in a controlled workspace



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Platform



Langfuse

Every assisted turn is traceable end to end

langfuse v3.178.1 OSS

PIX BCG / PIX Project Optimizer

Tracing

Traces Observations

Hide filters Views Search... IDs / Names 90d Past 90 days Off Columns 39/39

Filters

Environment

Mode SELECT TEXT

test default Only

Trace Name

Trace ID

User ID

Session ID

Metadata

Version

Release

Bookmarked

Comment Count

Comment Content

Tags

Level

Latency

Input Tokens

Output Tokens

Total Tokens

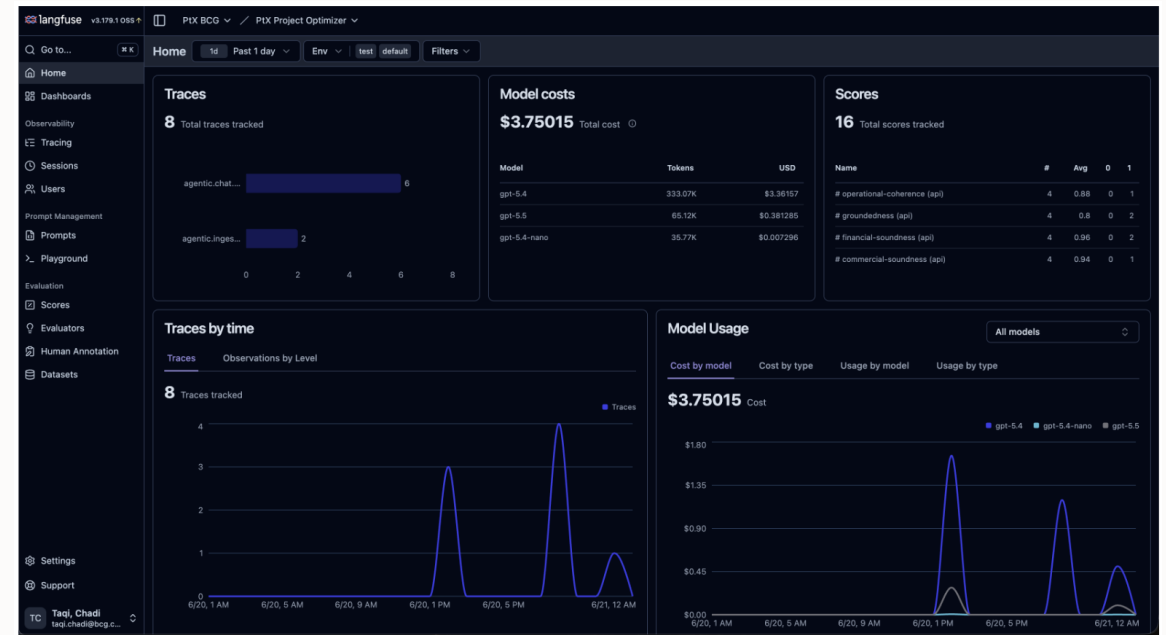
Input Cost

Output Cost

Total Cost

Input	Output	Observation Levels	Latency	Tokens
{ "chars": 102, "stage": "input" }	{ "answer": "...No battery is installed", so this run is not earning thr...	35	22.75s	138,872 → 2,228 (1 74,879)
{ "task_id": "8150b82aead745019b01c24a6364f012", "batch..." }	{ "task_id": "8150b82aead745019b01c24a6364f012", "batch..." }	10	1m 54s	
{ "task_id": "7dbab1e4eed54b5f8cccb8d41fe1d69ea", "batch..." }	{ "task_id": "7dbab1e4eed54b5f8cccb8d41fe1d69ea", "batch..." }	10	1m 55s	
{ "chars": 24, "stage": "input" }	{ "answer": "Opening the file upload for **may 2026 project** Once t...	25	5.66s	159,408 → 134 (1 79,771)
{ "chars": 11, "stage": "pre_right" }	{ "answer": "Hi. What are we tackling today?" }	14	2.23s	79,468 → 24 (1 39,749)
{ "chars": 102, "stage": "input" }	{ "answer": "...Revenue comes from ammonia offtake, not spot arbit..." }	35	17.12s	150,096 → 1,854 (1 80,404)
{ "chars": 778, "stage": "input" }	{ "answer": "...The monthly energy mix is active in **all 12 months** a..." }	35	14.35s	148,023 → 1,656 (1 79,341)
{ "chars": 889, "stage": "pre_right" }	{ "answer": "...The monthly quantity view is **active in all 12 months**..." }	35	15.95s	148,803 → 1,784 (1 79,821)
{ "chars": 785, "stage": "pre_right" }	{ "answer": "...**Effective revenue is active in every month** across t..." }	35	14.08s	145,475 → 1,541 (1 78,016)
{ "chars": 102, "stage": "input" }	{ "answer": "...Revenue is product led, not grid arbitrage led** on t..." }	35	20.34s	139,758 → 2,024 (1 75,323)
{ "chars": 30, "stage": "input" }	{ "passed": true, "triggered": [], "fail_open": true }	20	3.01s	80,818 → 94 (1 40,450)
{ "chars": 11, "stage": "input" }	{ "answer": "Hi. What are we tackling today?" }	14	2.43s	80,738 → 44 (1 40,391)
{ "chars": 31, "stage": "pre_right" }	{ "passed": true, "triggered": [], "fail_open": true }	20	4.30s	79,540 → 90 (1 39,819)
{ "chars": 41, "stage": "input" }	{ "answer": "From the page context, this project is **Seligli Battery a..." }	28	15.82s	169,734 → 1,108 (1 85,421)
{ "chars": 31, "stage": "input" }	{ "passed": true, "triggered": [], "fail_open": true }	20	3.71s	79,420 → 66 (1 39,743)
{ "chars": 76, "stage": "input" }	{ "passed": true, "triggered": [], "fail_open": true }	1	8.32s	67,820 → 864 (1 38,767)
{ "chars": 788, "stage": "pre_right" }	{ "answer": "...The **Windmill load factor** view is a m..." }	37	16.66s	225,863 → 2,292 (1 118,59)
{ "chars": 788, "stage": "input" }	{ "passed": true, "triggered": [], "fail_open": true }	2	2.29s	8,979 → 48 (1 9,027)
{ "chars": 248, "stage": "input" }	{ "answer": "...The scatter shows a **perfect positive c..." }	36	14.98s	209,548 → 2,091 (1 110,27)
{ "chars": 100, "stage": "pre_right" }	{ "answer": "...There is no battery arbitrage in simulat..." }	42	24.82s	223,719 → 3,569 (1 118,09)
{ "chars": 866, "stage": "pre_right" }	{ "answer": "...This run has **no battery storage** so t..." }	36	13.08s	200,337 → 1,167 (1 105,28)
{ "chars": 31, "stage": "input" }	{ "answer": "...count": 40, "projects": [{ "id": "1", "name": "Demo ... }	23	7.93s	166,398 → 182 (1 83,290)
{ "chars": 27, "stage": "pre_right" }	{ "passed": true, "triggered": [], "fail_open": true }	28	41.70s	107,679 → 1,996 (1 56,820)

Rows per page 50 Page 1 of 8



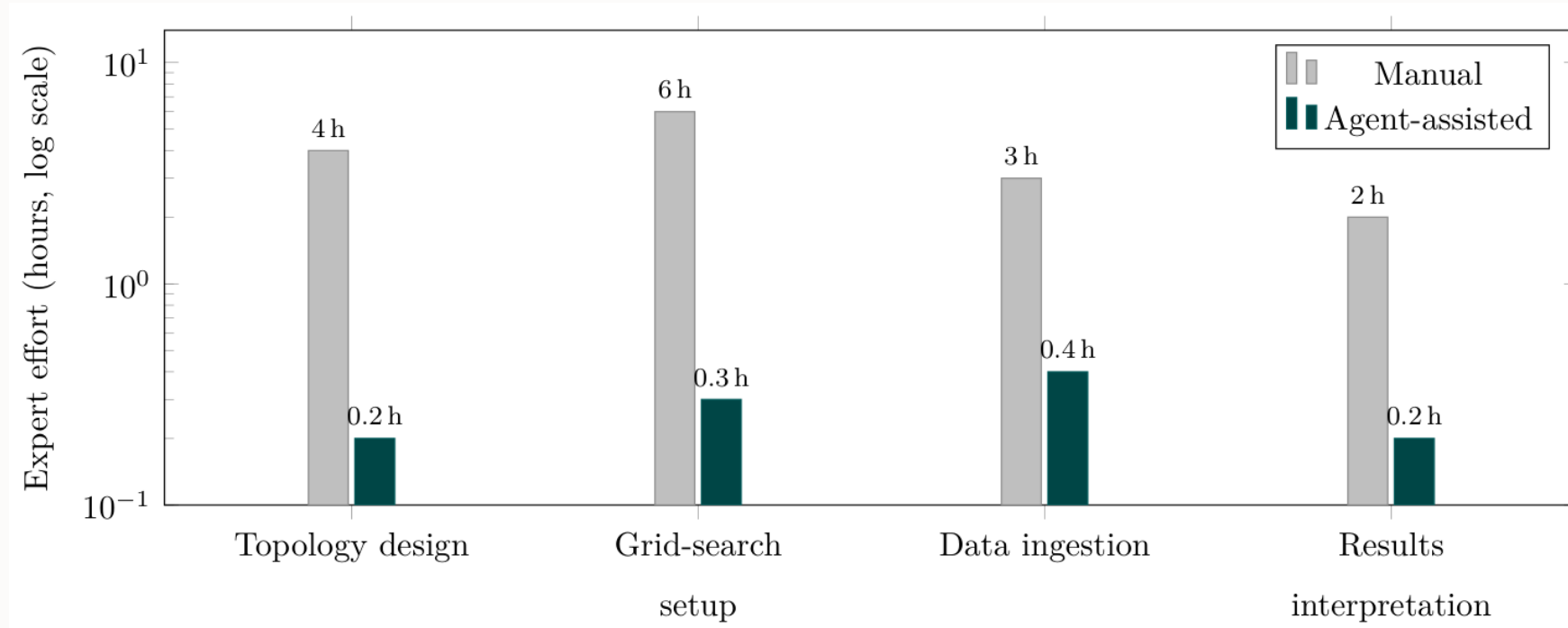
Langfuse traces (model calls, tools, costs, scores) and the versioned prompt registry

Section 6

Results

Measured, not assumed

Result 2: expert effort drops by an order of magnitude



- **Headline:** result interpretation went from about 2 hours of expert work to about 12 minutes of grounded conversation.
- **Honest framing:** first-order figures on a small sample, indicative rather than controlled.
- **The defensible gains:** consistency, traceability, and accessibility for non-specialists.

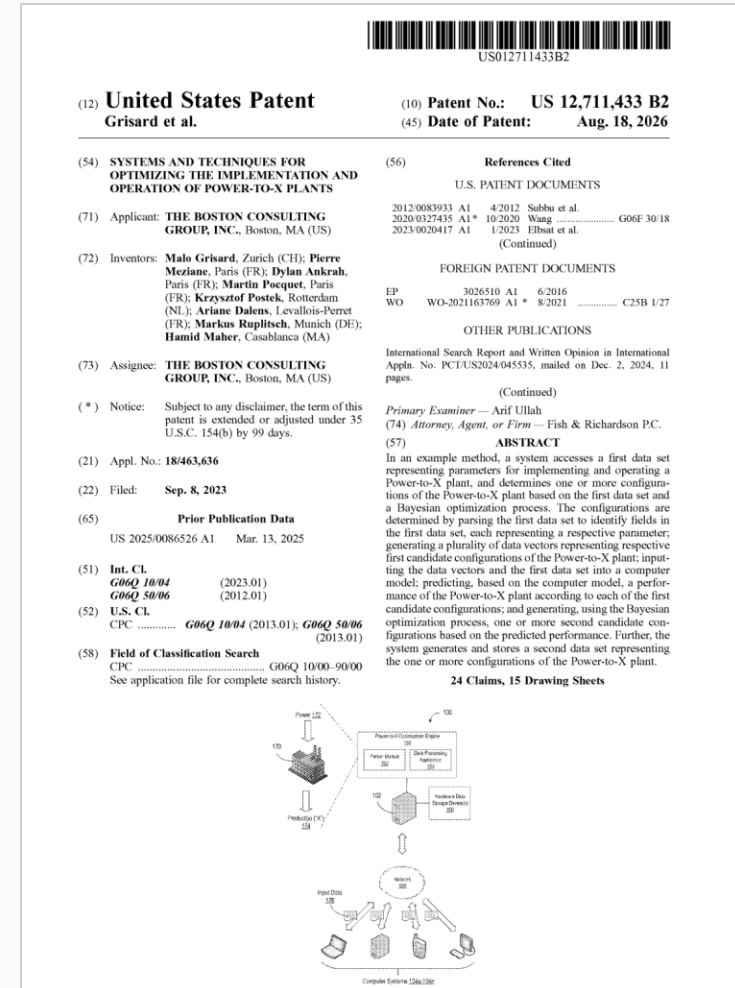
Section 7

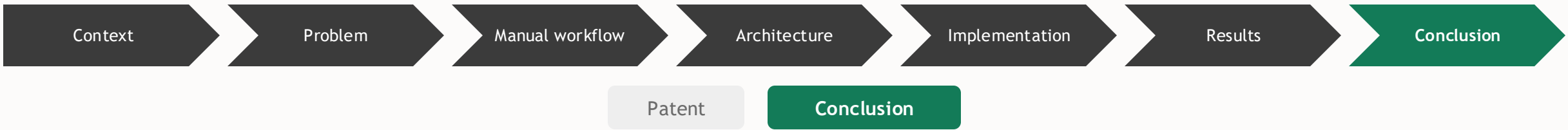
Conclusion

Recognition, the answer, and what comes next

Industrial recognition: a granted patent

- **US 12,711,433 B2**, granted to BCG on August 18, 2026, while this work was under way.
- It covers the **deterministic core's optimization method** (Bayesian candidate search over deterministic evaluation).
- **Why it matters here:** it certifies the value of exactly the numerical authority the governed layer is designed to serve, not replace.





The question, answered

How can generative AI make an expert-only optimizer accessible without compromising reproducibility, auditability, and human control?

By separating assistance from authority.

Five components, delivered end to end



Delivered

- Restored, reproducible platform foundation
- Governed multi-agent layer
- Grounded, read-only result interpreter
- Interactive results-visualization layer
- LLM-as-a-judge evaluation harness

Thank you

Any questions?

A Governed Multi-Agent Layer for a Power-to-X Pre-Feasibility Optimizer

Presented by:

EL OUKILI Tarik

Defended on **September 28, 2026**, in front of the jury composed of:

Pr. TAMTAOUI Ahmed

Pr. TOUNSI Karima

Pr. BAINA Amine

Mr. GRISARD Malo

Mr. MARTIN Paul

Mr. EL MAJIDI Hassan

- INPT (Examiner)

- INPT (Examiner)

- INPT (Internal Supervisor)

- BCG (Team Lead and Reviewer)

- BCG (Senior Team Member and Reviewer)

- BCG (External Supervisor)

